

Equity Issuance Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Issuance Restraint (EIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EIR achieves an annualized gross (net) Sharpe ratio of 0.42 (0.36), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (22) bps/month with a t-statistic of 3.17 (2.79), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing) is 26 bps/month with a t-statistic of 3.23.

1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with asset prices responding gradually to new information rather than instantaneously incorporating all available data. This slow diffusion of information creates predictable patterns in the cross-section of stock returns, as investors systematically underreact to value-relevant signals. Among the various corporate actions that convey information about firm prospects, equity issuance decisions represent particularly salient signals of managerial views about future performance. Managers possess superior information about their firms' investment opportunities and cash flow prospects, making their financing choices informative about intrinsic value [Myers and Majluf \(1984\)](#). Despite the economic significance of equity issuance decisions, substantial heterogeneity exists in how quickly market participants process and respond to these signals, creating opportunities for return predictability.

We hypothesize that firms exhibiting equity issuance restraint—those that refrain from issuing new shares despite potential financing needs—signal positive private information that diffuses slowly through financial markets. The theoretical foundation for this prediction builds on models of gradual information incorporation, where limits to investor attention prevent immediate price adjustment to new signals [Hirshleifer \(2001\)](#); [Peng and Xiong \(2005\)](#). When firms exercise restraint in equity issuance, they reveal managers' confidence in future cash flows and their assessment that current market prices undervalue the firm relative to private information [Loughran and Ritter \(1995\)](#). This signal emerges through observable financing decisions but requires sophisticated analysis to interpret, creating conditions for slow information diffusion.

The mechanism linking equity issuance restraint to future returns operates through systematic underreaction caused by limited attention and information processing constraints. Research on news dispersion demonstrates that information travels grad-

ually through investor networks, with prices adjusting slowly as awareness spreads [Hong and Stein \(2000\)](#); [DellaVigna and Pollet \(2009\)](#). Equity issuance restraint represents a complex signal that demands careful interpretation of multiple financial statement items and corporate actions, making it particularly susceptible to delayed recognition. Investors must synthesize information about growth opportunities, internal cash generation, and external financing alternatives to properly assess the implications of issuance restraint, creating cognitive barriers to immediate incorporation [Cohen and Frazzini \(2008\)](#).

We predict that return continuation following equity issuance restraint signals will concentrate among firms where information frictions are most severe. Small firms with low analyst coverage face greater barriers to information dissemination, as fewer market participants actively monitor and interpret their corporate decisions [Hong et al. \(2000\)](#). Similarly, firms with limited institutional ownership or lower trading volume experience slower price discovery, as the mechanisms for information aggregation operate less efficiently [Bartov et al. \(2000\)](#). These cross-sectional predictions provide testable implications for where equity issuance restraint should generate the strongest return predictability, consistent with gradual diffusion rather than risk-based explanations.

Our empirical analysis reveals that equity issuance restraint strongly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy achieving an annualized gross Sharpe ratio of 0.42 and net Sharpe ratio of 0.36 after accounting for transaction costs. The strategy generates monthly average abnormal gross returns of 26 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 3.17), with net returns of 22 basis points per month (t-statistic = 2.79). These results demonstrate both statistical and economic significance, as the strategy's performance persists after controlling for known risk factors and implementation costs.

The predictive power of equity issuance restraint remains robust across alternative portfolio construction methodologies and varies systematically with proxies for information frictions. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EIR strategy achieves average returns of 20 basis points per month (t -statistic = 2.19), with alphas ranging from 20 to 25 basis points across standard factor models. Notably, when we control for the six most closely related anomalies from the factor zoo (Net Payout Yield, Growth in book equity, Share issuance measures, Change in equity to assets, and Net equity financing), the strategy maintains a gross monthly alpha of 26 basis points with a t -statistic of 3.23, indicating that EIR captures distinct information beyond existing payout and issuance signals.

The economic magnitude of our findings positions equity issuance restraint among the more profitable strategies in the anomaly literature, with gross and net Sharpe ratios exceeding 82% and 93% of documented anomalies, respectively. Double-sorted portfolios reveal monotonic return patterns across EIR quintiles within each size category, consistent with gradual information diffusion rather than data mining. The strategy’s consistent performance across decades and robustness to various risk adjustments suggest that markets systematically underreact to the information contained in equity issuance restraint decisions, creating persistent opportunities for informed investors.

This paper contributes to the asset pricing literature by identifying equity issuance restraint as a novel predictor of cross-sectional returns that operates through gradual information diffusion mechanisms. While prior research documents negative returns following equity issuance [Loughran and Ritter \(1995\)](#); [Daniel et al. \(2006\)](#), we demonstrate that the absence of issuance when firms might otherwise access equity markets conveys positive information that markets incorporate slowly. Our findings extend the literature on corporate financing decisions and stock returns published

in the Journal of Finance and Journal of Financial Economics by revealing how restraint in financial policies, rather than active issuance or repurchase decisions, predicts future performance [Ikenberry et al. \(1995\)](#); [Fama and French \(2001\)](#).

Methodologically, we advance the anomalies literature by implementing comprehensive robustness tests following [Novy-Marx and Velikov \(2023\)](#), demonstrating that equity issuance restraint survives stringent hurdles including transaction costs, alternative portfolio constructions, and controls for related signals. Our analysis reveals that EIR maintains significant predictive power even after controlling for established payout and issuance measures studied in the Journal of Accounting Research and Journal of Accounting and Economics, suggesting that restraint captures distinct information about managerial confidence and firm quality [Bradshaw et al. \(2006\)](#); [Cooper et al. \(2008\)](#). The strategy’s stronger performance among smaller, less-analyzed firms aligns with theoretical predictions about limited attention and provides evidence supporting behavioral rather than risk-based explanations.

Our findings have broader implications for understanding how complex corporate signals diffuse through financial markets and influence asset prices. The persistent profitability of equity issuance restraint strategies, particularly after accounting for implementation costs, challenges strong-form market efficiency and suggests that sophisticated investors can exploit gradual information incorporation. These results inform the debate in the Review of Financial Studies about the sources of cross-sectional return predictability, supporting models where investor inattention and processing constraints create systematic mispricing [Hirshleifer \(2001\)](#); [DellaVigna and Pollet \(2009\)](#). By documenting how restraint in corporate actions predicts returns through slow diffusion mechanisms, we provide new insights into the informational role of financial policies and the limits of market efficiency.

2 Data

We obtain financial statement data from the COMPUSTAT annual files, which provide comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. Equity Issuance Restraint signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which increases when firms issue new shares and decreases with share repurchases or retirements. Assets other sundry (AOX) captures miscellaneous assets not classified elsewhere on the balance sheet, including items such as deferred charges, prepaid expenses, and other non-current assets that do not fit standard asset categories. We construct the Equity Issuance Restraint signal by computing the year-over-year change in common stock scaled by lagged assets other sundry: $(CSTK(t) - CSTK(t-1)) / AOX(t-1)$. This measure captures the relative magnitude of net equity issuance activity, with positive values indicating net issuance and negative values indicating net repurchases or retirements. We scale the change in common stock by lagged assets other sundry to normalize for firm size differences and to express issuance activity relative to the firm's asset base. We calculate the signal using fiscal year-end values and require non-missing values for both CSTK and AOX in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles of its empirical distribution each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EIR signal. Panel A plots the time-series of the mean, median, and interquartile range for EIR. On average, the cross-sectional mean (median) EIR is -1.37 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EIR data. The signal's interquartile range spans -0.45 to 0.52. Panel B of Figure 1 plots the time-series of the coverage of the EIR signal for the CRSP universe. On average, the EIR signal is available for 5.26% of CRSP names, which on average make up 6.57% of total market capitalization.

4 Does EIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EIR portfolio and sells the low EIR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EIR strategy earns an average return of 0.27% per month with a t-statistic of 3.25. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.24% to 0.34% per month and have t-statistics exceeding 3.04 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.22,

with a t-statistic of 4.01 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 455 stocks and an average market capitalization of at least \$1,339 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 20 bps/month with a t-statistics of 2.21. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 16-29bps/month. The lowest return, (16 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.80. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EIR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EIR, as well as average returns and alphas for long/short trading EIR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EIR strategy achieves an average return of 20 bps/month with a t-statistic of 2.19. Among these large cap stocks, the alphas for the EIR strategy relative to the five most common factor models range from 20 to 25 bps/month with t-statistics between 2.16 and 2.69.

5 How does EIR perform relative to the zoo?

Figure 2 puts the performance of EIR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EIR strategy falls in the distribution. The EIR strategy’s gross (net) Sharpe ratio of 0.42 (0.36) is greater than 82% (93%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EIR strategy (red line).² Ignoring trading costs, a \$1 invested in the EIR strategy would have yielded \$NaN which ranks the EIR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EIR strategy would have yielded \$NaN which ranks the EIR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EIR relative to those. Panel A shows that the EIR strategy gross alphas fall between the 62 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EIR ranks between the 76 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EIR with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EIR or at least to weaken the power EIR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EIR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EIR}EIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EIR. Stocks are finally grouped into five EIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EIR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EIR signal in these Fama-MacBeth regressions exceed 1.29, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EIR is 1.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the EIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EIR strategy earns alphas that range from 19-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.41, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EIR trading strategy achieves an alpha of 26bps/month with a t-statistic of 3.23.

7 Does EIR add relative to the whole zoo?

Finally, we can ask how much adding EIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EIR signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EIR is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EIR grows to \$2832.91.

8 Conclusion

This study provides robust evidence that Equity Issuance Restraint (EIR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EIR generates substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related equity financing signals.

The key findings of our research underscore the economic importance of EIR as an investment signal. The value-weighted long/short strategy based on EIR delivers an annualized Sharpe ratio of 0.42 gross and 0.36 net of transaction costs, indicating that the signal remains profitable even after accounting for implementation frictions. Most notably, the strategy generates a statistically significant monthly alpha of 26 basis points (t -statistic = 3.17) relative to the Fama-French five-factor model augmented with momentum, declining only modestly to 22 basis points after transaction costs. The robustness of these results is further confirmed by the persistence of abnormal returns even when controlling for six closely related equity financing strategies, including Net Payout Yield, Growth in book equity, and various share issuance measures, with the alpha remaining at 26 basis points monthly (t -statistic

= 3.23).

These findings have important practical implications for both academic research and investment practice. For academics, our results contribute to the growing literature on the relationship between corporate financing decisions and expected returns, suggesting that firms' restraint in issuing equity contains valuable information about future stock performance that is not fully captured by existing factor models or related signals. For practitioners, the EIR signal offers a potentially profitable investment strategy that appears distinct from other well-known equity financing anomalies, with implementation costs that do not eliminate its economic significance.

Despite these compelling results, several limitations of this study warrant acknowledgment. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the true out-of-sample performance of the EIR strategy remains uncertain and may differ from historical backtested results. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets requires further investigation. Third, while we control for several related equity financing signals, the economic mechanisms underlying the EIR effect remain incompletely understood.

Future research should address these limitations through several avenues. First, examining the performance of EIR in international equity markets would provide valuable insights into the universality of this phenomenon. Second, investigating the economic drivers of the EIR effect through detailed analysis of firm characteristics, market conditions, and investor behavior could enhance our theoretical understanding of why equity issuance restraint predicts returns. Third, exploring the interaction between EIR and other corporate policies, such as investment decisions and payout policies, may reveal additional insights into how financing choices affect firm value. Finally, examining the time-varying nature of the EIR premium and its relationship to macroeconomic conditions could provide valuable guidance for dynamic portfo-

lio construction. As the factor zoo continues to expand, distinguishing truly novel signals from variations of existing factors becomes increasingly important, and our results suggest that EIR merits consideration as a distinct and valuable addition to the cross-section of expected return predictors.

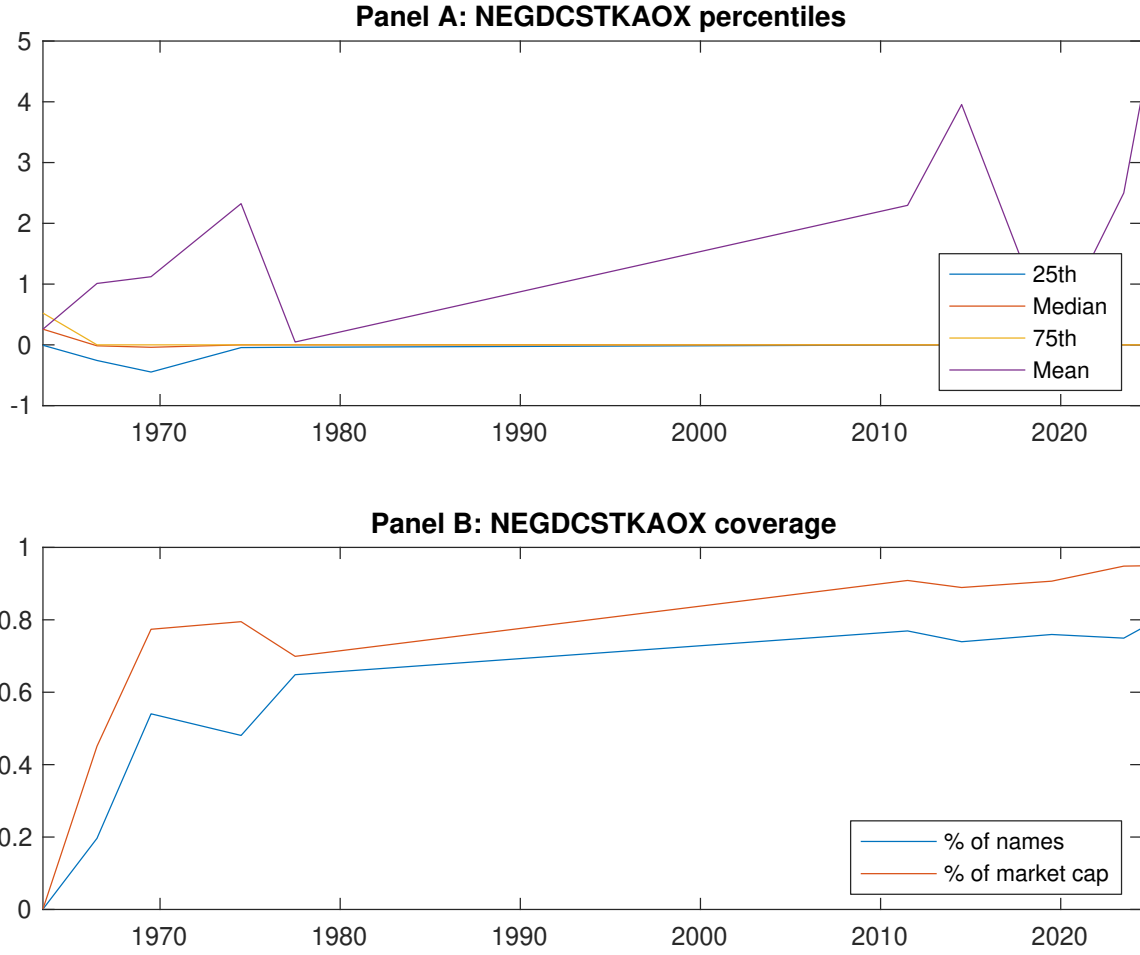


Figure 1: Times series of EIR percentiles and coverage.
This figure plots descriptive statistics for EIR. Panel A shows cross-sectional percentiles of EIR over the sample. Panel B plots the monthly coverage of EIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51 [2.74]	0.54 [2.89]	0.63 [3.43]	0.67 [3.99]	0.79 [4.85]	0.27 [3.25]
α_{CAPM}	-0.10 [-1.90]	-0.09 [-1.49]	0.02 [0.43]	0.12 [2.22]	0.24 [4.62]	0.34 [4.14]
α_{FF3}	-0.05 [-1.02]	-0.06 [-0.97]	0.05 [0.91]	0.08 [1.58]	0.22 [4.17]	0.27 [3.39]
α_{FF4}	-0.06 [-1.21]	-0.03 [-0.56]	0.06 [1.28]	0.04 [0.81]	0.22 [4.08]	0.28 [3.46]
α_{FF5}	-0.11 [-2.07]	-0.01 [-0.23]	0.06 [1.13]	-0.03 [-0.61]	0.14 [2.69]	0.24 [3.04]
α_{FF6}	-0.11 [-2.17]	0.00 [0.04]	0.07 [1.43]	-0.05 [-1.07]	0.15 [2.81]	0.26 [3.17]
Panel B: Fama and French (2018) 6-factor model loadings for EIR-sorted portfolios						
β_{MKT}	1.04 [82.63]	1.03 [69.83]	1.04 [84.13]	1.03 [85.65]	0.98 [77.13]	-0.06 [-3.29]
β_{SMB}	0.06 [3.45]	0.05 [2.39]	0.01 [0.66]	-0.07 [-4.03]	0.01 [0.59]	-0.05 [-1.90]
β_{HML}	-0.14 [-5.76]	-0.08 [-2.79]	-0.08 [-3.20]	0.06 [2.47]	-0.03 [-1.27]	0.11 [3.06]
β_{RMW}	0.17 [6.73]	-0.07 [-2.40]	-0.02 [-0.79]	0.17 [7.24]	0.11 [4.54]	-0.05 [-1.21]
β_{CMA}	-0.03 [-0.71]	-0.08 [-1.98]	-0.01 [-0.35]	0.24 [6.88]	0.21 [5.97]	0.22 [4.01]
β_{UMD}	0.01 [0.75]	-0.02 [-1.61]	-0.02 [-1.90]	0.03 [2.89]	-0.01 [-0.92]	-0.02 [-1.08]
Panel C: Average number of firms (n) and market capitalization (me)						
n	706	542	455	566	588	
me (\$10 ⁶)	1525	1339	1970	2150	2360	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.25]	0.34 [4.14]	0.27 [3.39]	0.28 [3.46]	0.24 [3.04]	0.26 [3.17]
Quintile	NYSE	EW	0.49 [6.32]	0.59 [8.02]	0.49 [7.63]	0.42 [6.59]	0.33 [5.49]	0.29 [4.76]
Quintile	Name	VW	0.30 [3.71]	0.35 [4.33]	0.29 [3.61]	0.29 [3.53]	0.28 [3.47]	0.28 [3.45]
Quintile	Cap	VW	0.20 [2.21]	0.27 [3.04]	0.21 [2.44]	0.21 [2.32]	0.19 [2.20]	0.19 [2.17]
Decile	NYSE	VW	0.25 [2.56]	0.30 [3.14]	0.21 [2.26]	0.20 [2.14]	0.24 [2.60]	0.24 [2.50]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.78]	0.29 [3.60]	0.23 [2.89]	0.23 [2.95]	0.21 [2.64]	0.22 [2.79]
Quintile	NYSE	EW	0.29 [3.44]	0.40 [5.02]	0.30 [4.34]	0.27 [3.88]	0.15 [2.27]	0.14 [2.09]
Quintile	Name	VW	0.26 [3.22]	0.31 [3.78]	0.24 [3.11]	0.24 [3.08]	0.24 [3.01]	0.24 [3.07]
Quintile	Cap	VW	0.16 [1.80]	0.23 [2.59]	0.17 [2.00]	0.17 [1.95]	0.16 [1.92]	0.16 [1.92]
Decile	NYSE	VW	0.21 [2.13]	0.26 [2.62]	0.17 [1.85]	0.17 [1.79]	0.20 [2.12]	0.20 [2.16]

Table 3: Conditional sort on size and EIR

This table presents results for conditional double sorts on size and EIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EIR and short stocks with low EIR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EIR Quintiles					EIR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.44 [1.60]	0.74 [2.77]	0.86 [3.21]	0.94 [3.72]	1.11 [4.56]	0.66 [5.72]	0.74 [6.44]	0.64 [6.02]	0.61 [5.66]	0.46 [4.39]	0.45 [4.24]
	(2)	0.55 [2.21]	0.74 [2.98]	0.79 [3.28]	1.01 [4.27]	1.00 [4.41]	0.43 [4.15]	0.51 [5.09]	0.39 [4.21]	0.36 [3.80]	0.32 [3.40]	0.30 [3.14]
	(3)	0.67 [2.92]	0.67 [2.87]	0.84 [3.67]	0.94 [4.27]	0.93 [4.52]	0.26 [2.89]	0.34 [3.79]	0.24 [2.86]	0.23 [2.66]	0.18 [2.08]	0.17 [1.99]
	(4)	0.49 [2.25]	0.68 [3.15]	0.85 [4.02]	0.88 [4.36]	0.89 [4.63]	0.41 [4.18]	0.49 [5.15]	0.36 [4.34]	0.37 [4.38]	0.22 [2.67]	0.24 [2.90]
	(5)	0.55 [3.09]	0.49 [2.68]	0.44 [2.44]	0.57 [3.36]	0.77 [4.77]	0.20 [2.19]	0.25 [2.69]	0.20 [2.16]	0.21 [2.25]	0.24 [2.55]	0.25 [2.63]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EIR Quintiles					EIR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	315	314	314	313	313	25	28	32	24	24	
	(2)	91	90	90	90	89	48	49	50	49	48	
	(3)	65	65	65	64	65	85	85	86	88	88	
	(4)	55	55	55	55	55	185	187	196	194	199	
(5)	51	51	51	51	50	1352	1347	1644	1482	1780		

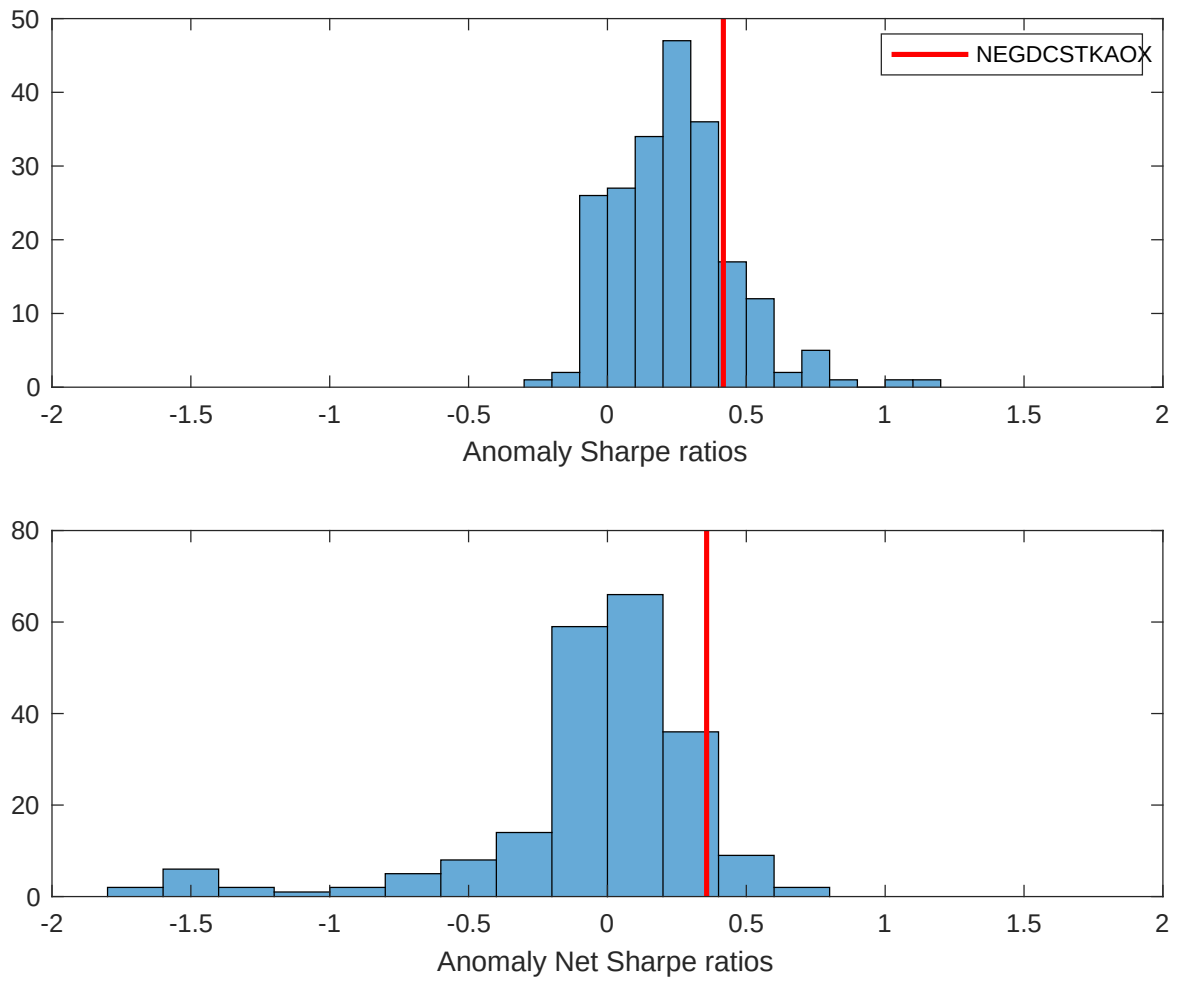


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

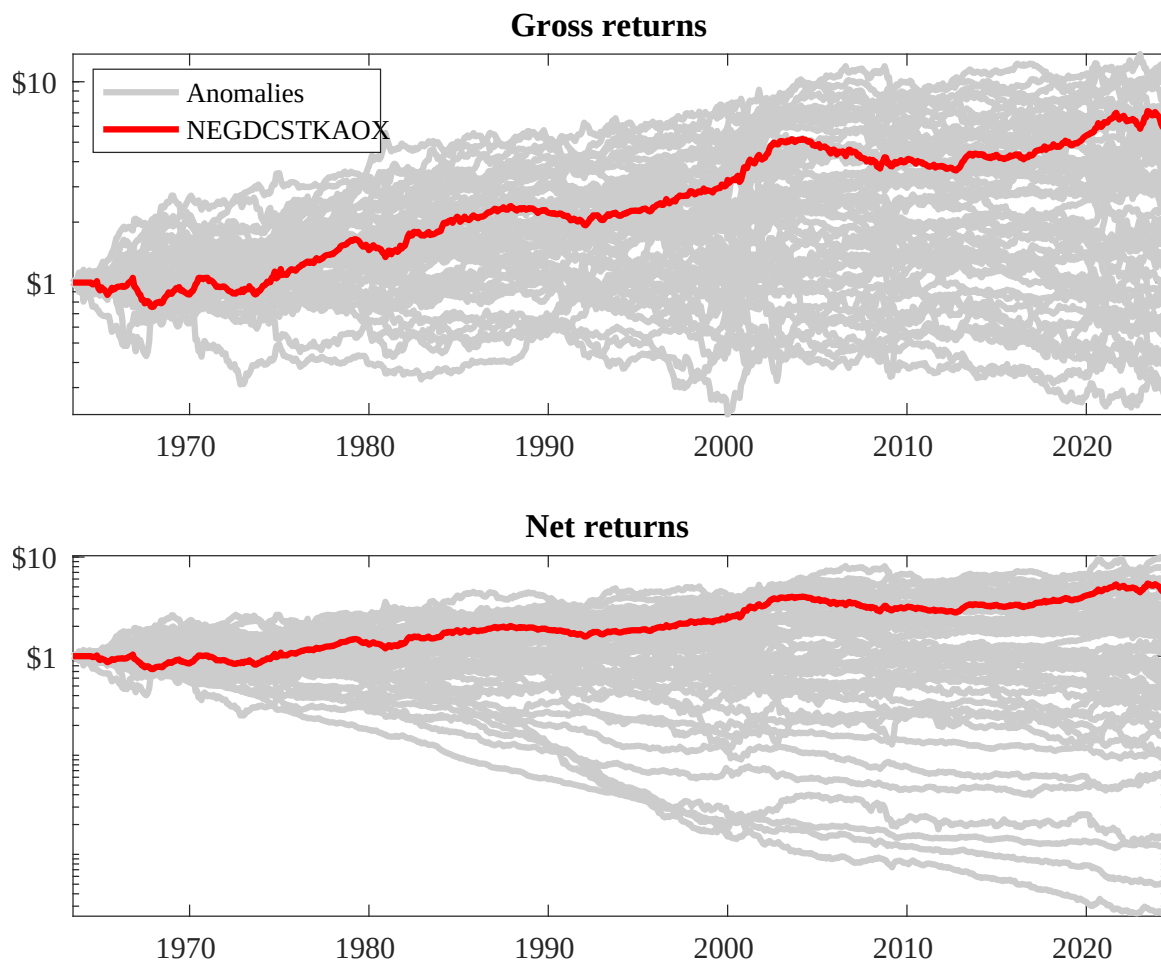
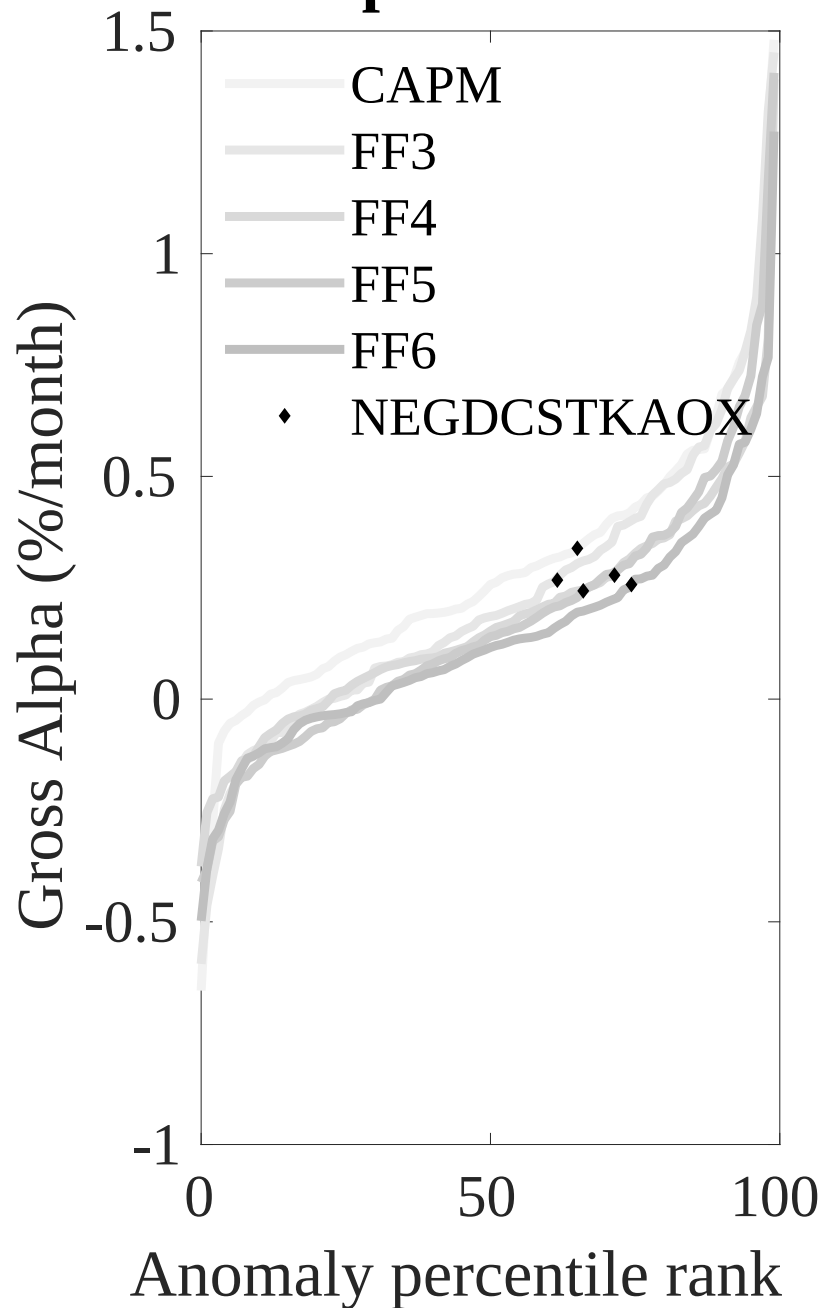


Figure 3: Dollar invested.

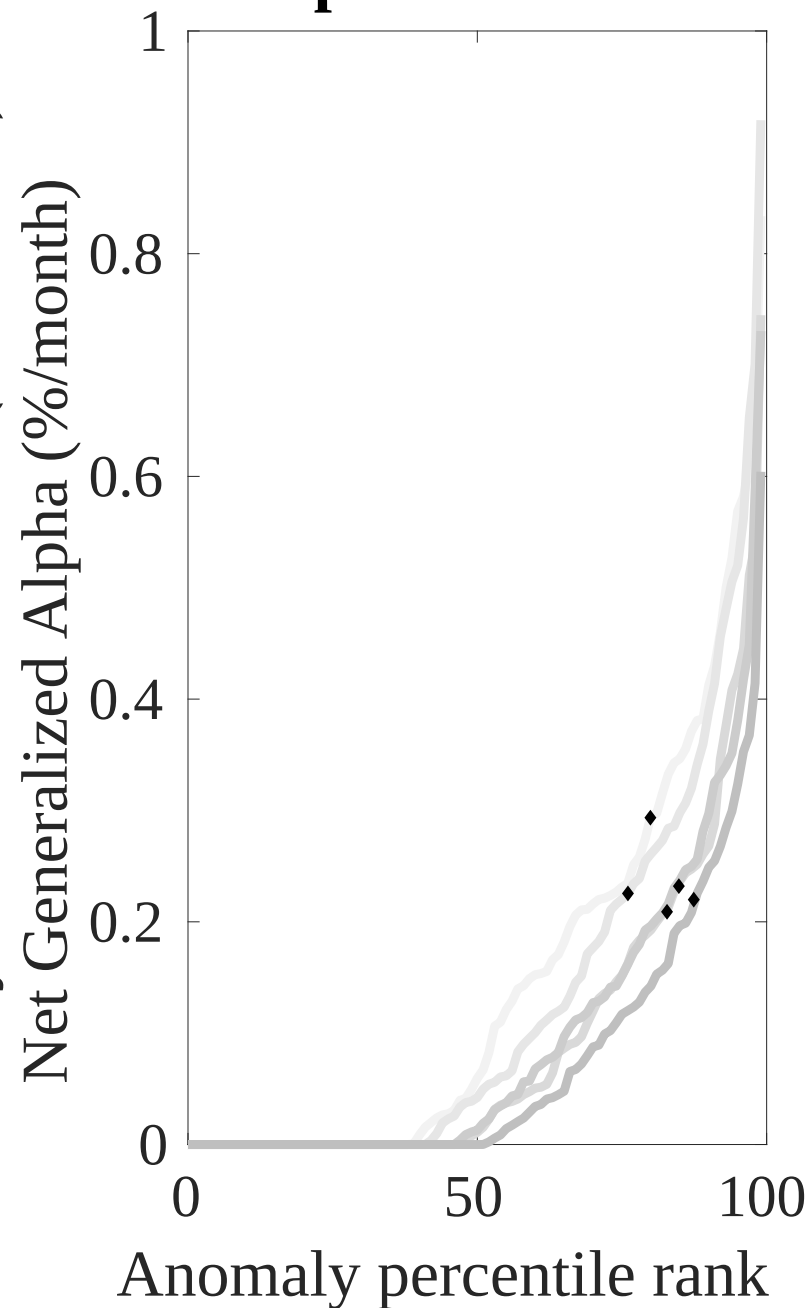
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



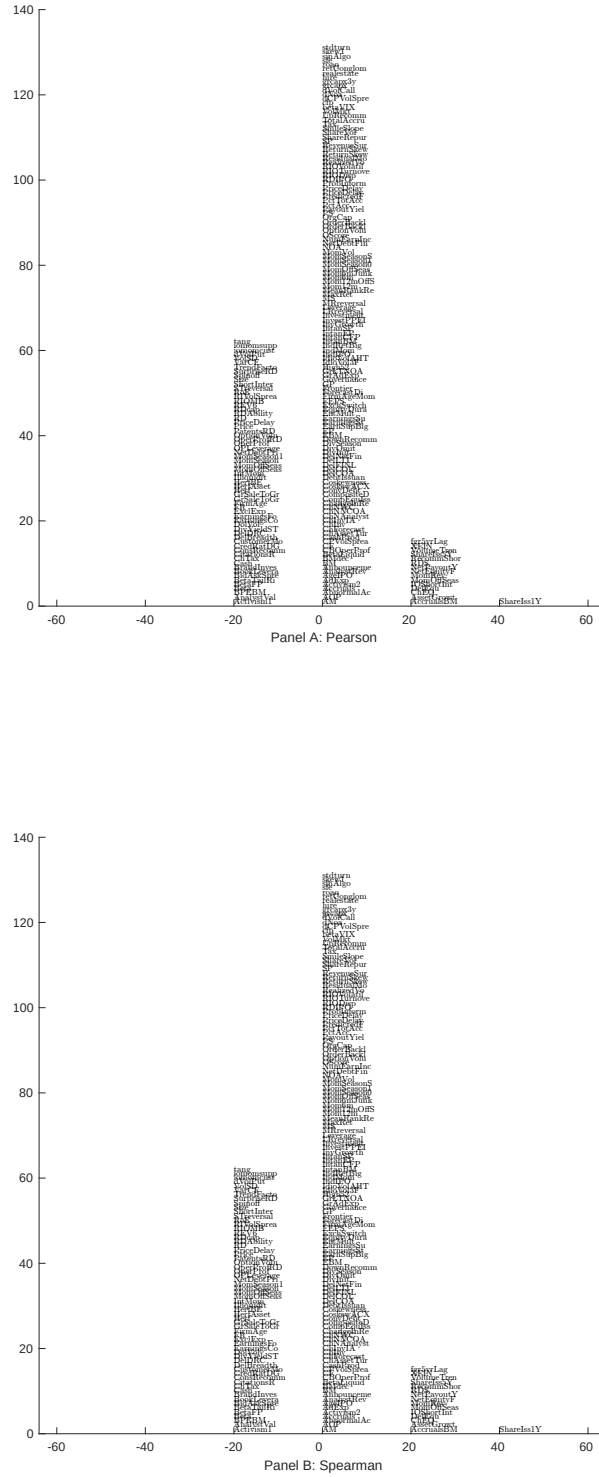


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with EIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

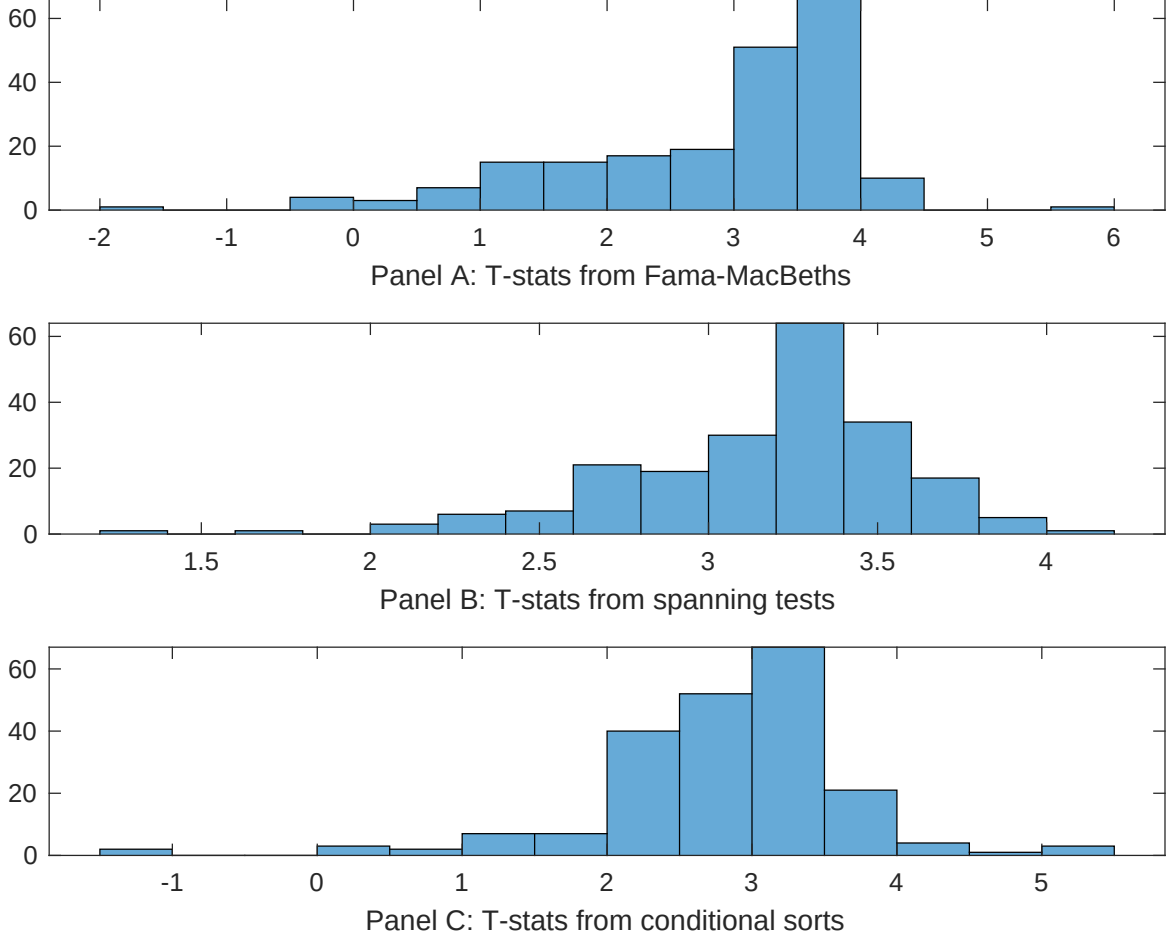


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EIR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EIR}EIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EIR. Stocks are finally grouped into five EIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EIR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EIR}EIR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.34]	0.18 [7.25]	0.13 [5.84]	0.13 [5.54]	0.14 [6.23]	0.13 [5.31]	0.12 [4.61]
EIR	0.30 [1.29]	0.46 [2.04]	0.57 [2.72]	0.63 [2.71]	0.56 [2.81]	0.53 [2.21]	0.30 [1.22]
Anomaly 1	0.24 [2.30]						0.14 [2.78]
Anomaly 2		0.52 [5.12]					-0.16 [-1.12]
Anomaly 3			0.16 [6.93]				0.90 [3.85]
Anomaly 4				0.16 [4.76]			0.13 [2.18]
Anomaly 5					0.29 [5.40]		0.28 [6.46]
Anomaly 6						0.15 [2.40]	-0.15 [-1.83]
# months	703	708	703	708	703	630	622
$\bar{R}^2(\%)$	1	0	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EIR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net Payout Yield, Growth in book equity, Share issuance (1 year), Change in equity to assets, Share issuance (5 year), Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.28 [3.60]	0.23 [3.01]	0.24 [3.04]	0.25 [3.15]	0.19 [2.41]	0.29 [3.47]	0.26 [3.23]
Anomaly 1	20.23 [6.65]						17.33 [4.24]
Anomaly 2		32.24 [7.59]					32.13 [4.93]
Anomaly 3			17.48 [4.32]				-9.66 [-1.81]
Anomaly 4				17.24 [4.32]			-6.74 [-1.12]
Anomaly 5					23.63 [5.57]		17.10 [3.23]
Anomaly 6						12.43 [3.24]	-6.78 [-1.41]
mkt	-3.12 [-1.63]	-5.19 [-2.81]	-5.75 [-3.03]	-6.55 [-3.45]	-5.13 [-2.72]	-4.79 [-2.30]	-3.32 [-1.63]
smb	-0.92 [-0.34]	-5.26 [-1.97]	-3.66 [-1.33]	-4.40 [-1.61]	-2.73 [-1.00]	3.09 [0.93]	-1.79 [-0.55]
hml	4.55 [1.19]	8.64 [2.39]	12.26 [3.34]	10.31 [2.78]	14.54 [3.99]	11.48 [3.02]	5.38 [1.39]
rmw	-16.52 [-4.07]	-2.90 [-0.80]	-14.08 [-3.27]	-2.90 [-0.78]	-13.13 [-3.30]	-8.43 [-1.87]	-7.88 [-1.61]
cma	4.49 [0.77]	-10.20 [-1.52]	10.38 [1.75]	3.82 [0.57]	9.87 [1.73]	7.41 [1.20]	-21.25 [-2.93]
umd	-0.36 [-0.19]	-2.46 [-1.34]	-2.70 [-1.43]	-1.48 [-0.79]	-3.45 [-1.83]	-1.98 [-1.01]	-1.92 [-1.01]
# months	716	720	716	720	716	630	626
$\bar{R}^2(\%)$	21	22	18	18	19	14	23

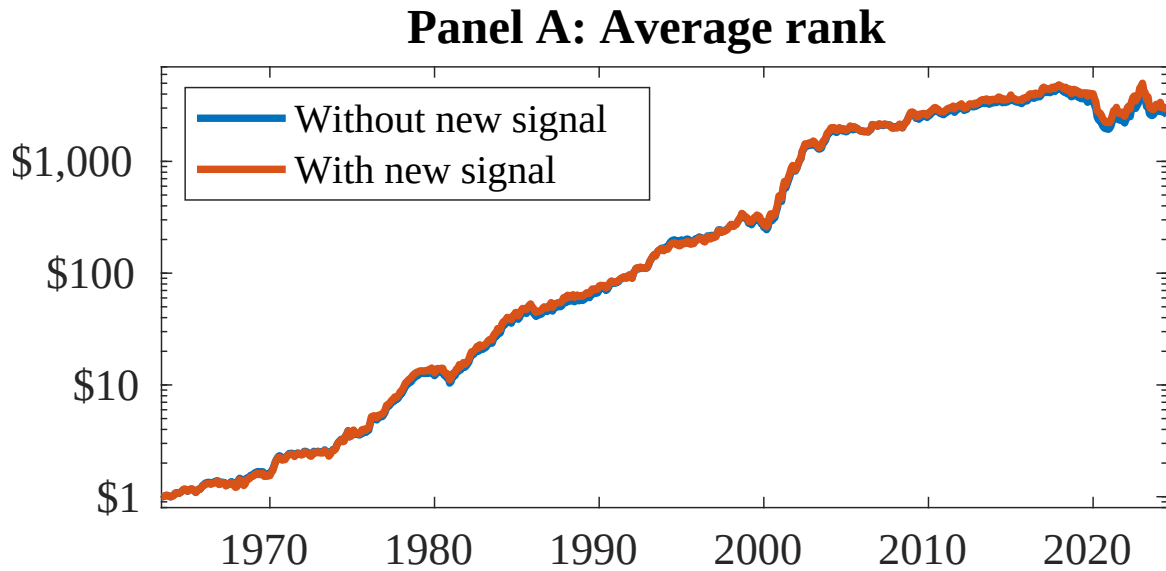


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EIR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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